

Exercise Sheet 4

No Fear of Numbers: Introduction to Quantitative Data Analysis in R

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Multivariate Analysis: Regression Models

So far, we have looked at our data **one variable at a time** (univariate) and in **pairs** (bivariate).

In this sheet, we take the next step to **multivariate analysis**: we use **regression models** to study how several predictors together are related to an outcome.

In the context of our research story, our main outcomes of interest are still:

- overall life satisfaction (bpsy01), and
- having children (yes/no, blcd06).

Now we want to answer questions like:

- How is life satisfaction related to **several factors at once** (e.g. work–life compatibility, emotional support, burden to parish, self efficacy, income)?
- Does a predictor still matter for life satisfaction **after controlling for** other variables?
- How do different predictors relate to the **probability of having children**?

In this exercise, you will learn to:

- fit and interpret a **multiple linear regression** with bpsy01 as the dependent variable,
- fit and interpret a **logistic regression** with blcd06 (children yes/no) as the dependent variable,
- report and interpret **regression coefficients, confidence intervals**, and (for logistic regression) **odds ratios**,
- summarize the main results in simple sentences (e.g. “Holding other variables constant, higher burden is associated with lower life satisfaction”).

0. Set-up: Load packages and dataset, load data frame

1. Install **tidyverse**, **haven**, **tidymodels** if not already installed and load these packages. Define the *exercise* folder as your working directory with the function `setwd("path")` and load dataset “04_qa_multi_data.sav” as a data frame called `mydata4`.

Code it (or faster: copy and paste the code) like this:

```

# Install tidyverse
if (!requireNamespace("tidyverse", quietly = TRUE))
  install.packages("tidyverse")

# Install tidymodels
if (!requireNamespace("tidymodels", quietly = TRUE))
  install.packages("tidymodels")

# Install haven
if (!requireNamespace("haven", quietly = TRUE))
  install.packages("haven")

## load packages
library(tidyverse)
library(haven)
library(tidymodels)

# set working directory
setwd(
  ↪ "C:/Users/Susanne/Nextcloud/share_DSC/003_Trainings/001_Workshops/2026/2026-03-11_SdV_MM_Quantitative_Data_Anal
  ↪ )

# Load the Nacaps data set and name this data frame "mydata4"
mydata4 <- read_sav("04_qa_multi_data.sav")

```

0 Prep: Converting all categorical to factor variables

In preparation of the multiple linear regression, make sure the categorical variables are factors (adbi15, adem01, adem03, apar10, bdbi01, blcd01, blcd06).

The code should look like this:

```

mydata4 <- mydata4 %>%
  mutate(
    across(
      c(adbi15, adem01, adem03, apar10, bdbi01, blcd01, blcd06), # all cat. variables
      ~ haven::as_factor(.x) # labels -> factor levels
    )
  )
glimpse(mydata4)

```

Exercise 1: Multiple linear regression: life satisfaction

We want to model **overall life satisfaction** (bpsy01) as a function of PhD characteristics and work conditions, attitudes and well-being, and control for demographics. We use **multiple linear regression** with `lm()` and **tidy the output** with *tidy*.

1.1. Create a cleaned data frame `mydata_lm_ex1` that only contains rows with non-missing values for the dependent variable and all predictors in this model.

The code should look like this:

```
mydata_lm_ex1 <- mydata4 %>%
  filter(
    !is.na(bpsy01),
    !is.na(bdbi01),
    !is.na(adbi15),
    !is.na(bdcd17),
    !is.na(bdwr12),
    !is.na(bemp81),
    !is.na(blcd12),
    !is.na(apar15b),
    !is.na(bpsy05),
    !is.na(adem01),
    !is.na(adem02),
    !is.na(apar10),
    !is.na(adem03),
    !is.na(blcd01),
    !is.na(blcd06)
  )
```

The new data frame `mydata_lm_ex1` should now appear in your environment pane.

1.2. Fit a multiple linear regression with `bpsy01` as the dependent variable and all other variables as predictors/controls.

The code should look like this:

```
lm_phd <- lm( # save results in data frame lm_phd
  bpsy01 ~ bdbi01 + adbi15 + bdcd17 + bdwr12 + bemp81 + #bpsy01 as dependent var, predictors: PhD
  ⇨ and work conditions
    blcd12 + apar15b + bpsy05 + # predictors: attitudes and
  ⇨ well-being
    adem01 + adem02 + apar10 + adem03 + blcd01 + blcd06, # controls: demographics
  data = mydata_lm_ex1 # use data frame mydata_lm_phd
)
```

Tipp: By default, R uses the first factor level (often the lowest numeric code) as the reference category. You can change the reference directly inside the regression call using `relevel()` in the formula. For example like this:

```
lm_phd <- lm(
  bpsy01 ~
    bdbi01 +
    relevel(adbi15, ref = "mathematics, natural sciences") + # change reference category here
    bdcd17 +
    bdwr12 +
    bemp81 +
    blcd12 +
    apar15b +
    bpsy05 +
    adem01 +
    adem02 +
    apar10 +
    adem03 +
    blcd01 +
    blcd06,
  data = mydata_lm_ex1
)
```

1.3. Create a tidy table of coefficients and check the results.

Do it like this:

```
# use tidy data frame lm_phd, save results in data frame lm_phd_results and display 95% confidence
  ↪ intervals
lm_phd_results <- tidy(lm_phd, conf.int = TRUE) %>%
  mutate(across(where(is.numeric), ~ round(.x, 3)) # optional: round output to three decimals
  )

print(lm_phd_results, n = Inf) # show all rows
```

```
## # A tibble: 26 x 7
##   term                estimate std.error statistic p.value conf.low conf.high
##   <chr>              <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
## 1 (Intercept)        3.01      0.479      6.27      0         2.07      3.95
## 2 bdbi01I have complet~ -0.168    0.12     -1.41     0.16    -0.403    0.066
## 3 bdbi01I have interrui~  0.132    0.238     0.556    0.578   -0.335    0.599
## 4 bdbi01I have quit my~ -0.423    0.414    -1.02     0.307   -1.24     0.389
## 5 adbi15sports       -0.175    0.398    -0.44     0.66    -0.955    0.605
## 6 adbi15law, economics~ -0.096    0.131    -0.734    0.463   -0.354    0.161
## 7 adbi15mathematics, n~ -0.03     0.127    -0.237    0.812   -0.28     0.219
## 8 adbi15human medicine~  0.269    0.151     1.79     0.074   -0.026    0.564
## 9 adbi15agricultural, ~ -0.043    0.255    -0.166    0.868   -0.543    0.458
## 10 adbi15engineering sc~ -0.344    0.145    -2.37     0.018   -0.628   -0.059
## 11 adbi15art, art theory  0.283    0.278     1.02     0.309   -0.262    0.828
## 12 bdcd17             0.188    0.032     5.93      0         0.126     0.25
## 13 bdwr12            -0.102    0.048    -2.12     0.034   -0.196   -0.008
## 14 bemp81              0         0         4.82      0         0         0
## 15 blcd12             0.332    0.017    19.6      0         0.298    0.365
## 16 apar15b            0.086    0.053     1.63     0.103   -0.017    0.19
## 17 bpsy05              0.645    0.06     10.7      0         0.527    0.763
## 18 adem01male         -0.08     0.081    -0.989    0.323   -0.24     0.079
## 19 adem01other         0.076    0.172     0.443    0.658   -0.262    0.414
## 20 adem02            -0.036    0.008    -4.76      0    -0.051   -0.021
## 21 apar10Bachelor / Mas~  0.174    0.112     1.55     0.121   -0.046    0.394
## 22 apar10Other vocation~  0.116    0.114     1.02     0.309   -0.107    0.339
## 23 apar10Unclassifiable  0.126    0.26     0.484    0.629   -0.385    0.636
## 24 adem03in another cou~ -0.093    0.107    -0.866    0.387   -0.304    0.118
## 25 blcd01no          -0.527    0.091    -5.78      0    -0.705   -0.348
## 26 blcd06no          -0.647    0.111    -5.84      0    -0.864   -0.43
```

You get the following information:

- *term*: variable name (intercept and predictors)
- *estimate*: regression coefficient (change in bpsy01 if the predictor increases by 1, holding - other variables constant; for factors: difference to the reference category, this is always the first)
- *std.error*, *statistic*, *p.value* = test statistics
- *conf.low*, *conf.high* = lower and upper bound of the 95% CI

How do you interpret the results?

Solution:

For **factors** (bdbi01, adbi15,...) the coefficients are differences in the mean compared to the reference category, while holding all other variables constant.

Describe it in the results section like this:

PhD status: Compared to currently enrolled PhD candidates, those who completed, interrupted, or quit their PhD do not differ significantly in overall life satisfaction, indicating that formal PhD status itself is not systematically associated with well-being in this model.

Field of study (reference: Humanities): Compared to respondents in humanities, only those in engineering sciences report significantly lower life satisfaction, while all other disciplinary groups do not differ significantly, suggesting largely similar well-being levels across fields.

For **metric** predictors (**blcd17**, **bdwr12**, **bemp81**,—), you see the slope per 1-unit (e.g. 1 Euro income or 1-point on likert scale) increase.

Describe it in the results section like this:

Emotional support during the PhD: For each one-point increase in perceived emotional support during the PhD, overall life satisfaction increases by about 0.19 points on average, and this positive association is highly statistically significant ($p < .001$).

1.4. Use the `glance()` function to get an overview of the model fit.

Code it like this:

```
lm_phd_fit <- glance(lm_phd)
lm_phd_fit
```

```
## # A tibble: 1 x 12
##   r.squared adj.r.squared sigma statistic  p.value    df logLik   AIC   BIC
##   <dbl>      <dbl> <dbl>      <dbl>    <dbl> <dbl> <dbl> <dbl> <dbl>
## 1    0.362      0.352  1.52      37.6 7.36e-142    25 -3083. 6220. 6366.
## # i 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>
```

Important values:

- *r.squared*: proportion of variance in life satisfaction explained by the model
- *adj.r.squared*: adjusted R^2 (corrected for number of predictors)
- *p.value* (for the overall F-test): tests if the model explains more variance than a model with no predictors*

How would you interpret the results?

Solution:

The regression model explains **about 36 % of the variance** in overall life satisfaction. The overall model is statistically significant.

Exercise 2: Multiple logistic regression: probability of having children

We want to predict the probability of **having children (yes/no)** (**blcd06**) as a function of PhD and work conditions, attitudes and well-being and demographics.

We estimate the model with the `glm()` function and use `tidy()` to get odds ratios.

2.1. Create a cleaned data frame `mydata_lm_ex2` that only contains rows with non-missing values for the dependent variable and all predictors in this model.

Solution:

```
mydata_glm_ex2 <- mydata4 %>%
  filter(
    !is.na(bpsy01),
    !is.na(bdbi01),
    !is.na(adbi15),
    !is.na(bdcd17),
    !is.na(bdwr12),
    !is.na(bemp81),
    !is.na(blcd12),
    !is.na(apar15b),
    !is.na(bpsy05),
    !is.na(adem01),
    !is.na(adem02),
    !is.na(apar10),
    !is.na(adem03),
    !is.na(blcd01),
    !is.na(blcd06)
  )
```

2.2. Fit the full logistic regression model on having children (yes/no) blcd06.

To model the log-odds of blcd06 for “yes” with reference category “no”, we have to relevel our dependent variable with `relevel()`.

The code goes like this:

```
glm_full <- glm(
  relevel(blcd06, ref = "no") ~                # change reference category
    bdbi01 + adbi15 + bdcd17 + bdwr12 + bemp81 + # predictors
    blcd12 + apar15b + bpsy05 + bpsy01 +
    adem01 + adem02 + apar10 + adem03 + blcd01,
  data = mydata_glm_ex2,                        # take dataset without missings
  family = binomial(link = "logit")             # logistic regression
)
```

2.3. Tidy the table and display odds ratios.

The code should look like this:

```
glm_ex2_results <- tidy(
  glm_full,
  exponentiate = TRUE, # exp(coef) = odds ratios instead of log-odds
  conf.int = TRUE # 95% confidence intervals
) %>%
  mutate(across(where(is.numeric), ~ round(.x, 3))) # round numeric values to three decimals for
  ↪ readability

print(glm_ex2_results, n = Inf) # show all rows
```

```
## # A tibble: 26 x 7
##   term                                estimate std.error statistic p.value conf.low conf.high
##   <chr>                                <dbl>     <dbl>     <dbl>   <dbl>   <dbl>   <dbl>
## 1 (Intercept)                        0         0.955     -8.25    0         0         0.002
## 2 bdbi01I have complet~             1.40      0.222      1.52   0.128     0.902     2.15
## 3 bdbi01I have interrui~            2.36      0.38       2.26   0.024     1.09      4.90
## 4 bdbi01I have quit my~            2.67      0.742      1.32   0.186     0.569    10.9
```

## 5	adbi15sports	1.74	0.724	0.761	0.446	0.353	6.54
## 6	adbi15law, economics~	1.04	0.249	0.149	0.881	0.638	1.70
## 7	adbi15mathematics, n~	0.829	0.252	-0.744	0.457	0.507	1.36
## 8	adbi15human medicine~	1.38	0.295	1.10	0.272	0.774	2.47
## 9	adbi15agricultural, ~	0.396	0.629	-1.47	0.141	0.1	1.23
## 10	adbi15engineering sc~	1.46	0.277	1.37	0.17	0.851	2.52
## 11	adbi15art, art theory	1.58	0.45	1.02	0.306	0.644	3.79
## 12	bdc17	1.12	0.064	1.71	0.087	0.985	1.27
## 13	bdwr12	0.982	0.093	-0.191	0.848	0.817	1.18
## 14	bemp81	1	0	-2.41	0.016	1	1
## 15	blcd12	0.857	0.037	-4.15	0	0.796	0.922
## 16	apar15b	1.05	0.108	0.441	0.659	0.85	1.30
## 17	bpsy05	0.831	0.129	-1.44	0.15	0.646	1.07
## 18	bpsy01	1.38	0.056	5.80	0	1.24	1.54
## 19	adem01male	1.05	0.165	0.308	0.758	0.762	1.45
## 20	adem01other	1.06	0.335	0.186	0.852	0.536	2.01
## 21	adem02	1.21	0.016	11.7	0	1.17	1.25
## 22	apar10Bachelor / Mas~	0.677	0.217	-1.8	0.072	0.445	1.04
## 23	apar100ther vocation~	0.505	0.227	-3.01	0.003	0.325	0.791
## 24	apar10Unclassifiable	0.702	0.497	-0.711	0.477	0.252	1.80
## 25	adem03in another cou~	0.999	0.213	-0.004	0.997	0.652	1.51
## 26	blcd01no	0.148	0.292	-6.55	0	0.08	0.254

In the output table, estimate now shows **odds ratios** (because we used `exponentiate = TRUE`).

Odds ratio > 1: the predictor is associated with higher odds of having children (`blcd06 = "yes"`). **Odds ratio < 1:** the predictor is associated with lower odds of having children.

The **95% confidence interval** shows the uncertainty around each odds ratio; if it includes 1, the effect is not statistically significant at the 5% level.

How do you interpret the results?

Solution:

For **categorical predictors**, the odds ratio (OR) indicates how the odds of the outcome differ between a given category and the reference category, holding all other variables constant.

Describe it in the results section like this:

Compared to currently enrolled PhD candidates (`bdbi01`), those who interrupted their PhD have significantly higher odds of having children ($OR = 2.36, p = .024$), meaning their odds are about 136% higher, whereas those who completed or quit their PhD do not differ significantly from the reference group.

For **metric predictors**, the odds ratio (OR) represents the multiplicative change in the odds of the outcome associated with a one-unit increase in the predictor, controlling for all other variables.

Describe it in the results section like this:

Each additional year of age (`adem02`) is associated with roughly 21% higher odds ($OR = 1.21, p < .001$) of having children.

2.4. Finally, calculate the overall model fit including McFadden R^2 .

The code should look like this:

```
glm_ex2_fit <- glance(glm_full)

glm_ex2_fit %>%
  mutate(
```

```
mcfadden_r2 = 1 - deviance / null.deviance
)
```

```
## # A tibble: 1 x 9
##   null.deviance df.null logLik   AIC   BIC deviance df.residual  nobs
##         <dbl>   <int>  <dbl> <dbl> <dbl>   <dbl>       <int> <int>
## 1         1518.    1681  -586. 1224. 1365.    1172.       1656  1682
## # i 1 more variable: mcfadden_r2 <dbl>
```

How do you interpret the model fit?

Solution:

The full logistic regression model has a McFadden pseudo- R^2 of about 0.23. Values between 0.2 and 0.4 are often interpreted as a good model fit in logistic regression. This means that our model substantially improves the prediction of who has children compared to an intercept-only model, even though there may be still a lot of unexplained variation.

Take Home checklist: Multivariate regression in R

#	Question / task	Useful functions / tools
1	What is your outcome variable and which type of regression do you need?	Check scale & distribution with <code>str()</code> , <code>summary()</code> , <code>frq()</code> , <code>skimr::skim()</code> , histograms / bar charts via <code>ggplot()</code>
2	Are predictors coded correctly (metric vs. factor, reference categories, NAs)?	Recode with <code>mutate()</code> , <code>across()</code> , <code>case_when()</code> , set NAs; convert labelled vars with <code>haven::as_factor()</code> , set factor order / reference with <code>forcats::fct_relevel()</code> or <code>relevel()</code>
3	Fit a model with a small set of predictors	Linear: <code>lm(outcome ~ x1 + x2, data = ...)</code> ; Logistic: <code>glm(I(outcome == "yes") ~ ..., family = binomial)</code> ; get coefficients with <code>broom::tidy(model, conf.int = TRUE)</code> ; model fit with <code>broom::glance(model)</code>
4	How do you interpret coefficients ?	Linear: change in outcome per 1-unit change in predictor, “holding other variables constant”; Logistic: use <code>tidy(glm_model, exponentiate = TRUE, conf.int = TRUE)</code> for odds ratios and 95% CIs, check if CI includes 1
5	How well does the model fit the data?	For linear: use <code>glance()</code> → <code>r.squared</code> , <code>adj.r.squared</code> ; for logistic: <code>glance()</code> → McFadden pseudo-R ² : <code>mutate(mcfadden_r2 = 1 - deviance / null.deviance)</code>
6	Can you communicate the results in simple sentences?	From <code>tidy()</code> and <code>glance()</code> : report direction and size of effects, p-values and confidence intervals, plus overall fit (R ² / pseudo-R ²); use phrases like “holding other variables constant...” and focus on substantive importance, not only significance